Module 2: Descriptive Statistics Review

Bring the question, data, tools, and interpretation together

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Purpose and submission

Prepare a **short Quarto report** using the checked textbook CSVs. This review brings Lessons 1–4 together. You should demonstrate that you can choose and interpret a tool, rather than simply produce every available statistic.

Include one concise section for each task below, with readable tables, labeled plots, and reproducible calculations. Read the data guide and use the CSV bundle. Keep source observations unchanged. Use type-6 quartiles consistently when requested.

These are the same business cases introduced in the lessons. Refer back to their English and Arabic introductions for context. Statistical explanations and answers remain in English.

1. Define the variables and choose tools

For Restaurant quality rating and meal price, Audit duration, SoftDrink brand preference, and Electronics commercials and sales:

1. Identify the observational unit, variable type, and measurement level. Distinguish numerical counts from measured quantities where relevant.
2. Choose a suitable table, chart, or numerical measure for each task below. Justify your choice in one sentence.
3. Explain why an average restaurant ID or average brand code would not answer a meaningful quantitative question.

2. Summarize one categorical distribution

Using **SoftDrink**:

1. Construct a frequency and percentage table using `brand_book_label`.
2. Draw a bar chart and identify the most frequent category.

3. State what one record represents. Explain why percentages here describe the recorded purchases rather than all consumers in a market.

3. Describe a numerical distribution

Using **Audit**:

1. Draw a histogram and a dot plot. Briefly discuss shape, center, spread, and unusual features. Explain why 20 observations and the histogram's bin width limit fine claims about modality.
2. Report n, mean, median, range, sample SD, type-6 quartiles, and IQR, with units.
3. Identify the 1.5-IQR fences and any flagged observations. Explain why a flag does not automatically justify deletion.
4. Recommend a center-and-spread pair for a typical-duration description, while explaining what useful information the other pair contributes.

4. Compare groups and percentages

Using **Restaurant**:

1. Show a compact table of group sizes, median prices, and type-6 IQRs. Compare groups using aligned boxplots with the same quartile convention.
2. Construct a crosstab with the four price bands from Lesson 4.
3. Compare the proportion priced at least USD 30 **within** each quality category. State the three denominators.
4. Explain why the crosstab and boxplots complement each other, and why these descriptive differences do not establish a causal effect of quality ratings on price.

5. Describe numerical association

Using **Electronics**:

1. Draw a scatterplot with clear sales units.
2. Report sample covariance and Pearson correlation. Explain why covariance changes when sales are converted to USD but correlation does not.
3. Write a concise conclusion supported by both the plot and the coefficient. Include a limitation concerning causal interpretation or generalization.

Review checklist

Criterion	Evidence in your report
Correct variable meaning	Observational units, labels, units, and classifications
Suitable tool selection	A reason for each summary or plot
Reproducible calculations	CSV input, correct denominators, stated quartile convention
Clear communication	Legible plots, concise tables, context-specific interpretations
Sound conclusions	Flags investigated; descriptive associations kept separate from causation

You are ready to move on when you can explain **why** each tool suits its question, describe what it leaves out, and recognize a conclusion that exceeds the evidence.

This completes the planned descriptive-statistics sequence. Module 3 now provides five lessons, practice, and a review on probability and distributions.

Return to the Module 2 overview.